

Fractional Order Bergman's Minimal Model—A Better Representation Of Blood Glucose-Insulin System

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Abstract—In this paper, the blood glucose-insulin system in the human body is discussed. Different mathematical models representing this system are presented. MATLAB is used for solution and simulation of each of these models. On the basis of credible experimental data, comparison of these models is done. The results show that the system of glucose-insulin is better described by nonlinear Bergman's model than the linear model. Furthermore, utilizing fractional calculus, integer order nonlinear Bergman's model is modified to fractional order model. It is shown through simulations and comparison of data that the modified fractional order nonlinear Bergman's minimal model gives better representation compared to the integer order nonlinear model.

Index Terms—blood glucose-insulin system, mathematical models, fractional calculus, Bergman's minimal model

I. INTRODUCTION

Glucose is the key source of energy required by the cells of living organisms to function. Range of glucose in the body of a normal healthy person is (70-110 mg/dL). After intake of meal, glucose concentration in blood increases, which is lowered down back to the normal value by a hormone released by pancreas called insulin [1]. Failure of pancreas to decrease glucose concentration to normal value is a disease called as Diabetes Mellitus.

Diabetes Mellitus is one of the most common diseases in the world. According to [2], in 2017, 425 million people of the age group 20-79 years, were suffering from this disease, and this number will rise to 629 million, by 2045. Furthermore, 4 million people have died because of this disease.

Mathematical modeling plays very important role in the representation of a phenomenon. On the basis of biological description, complexity, and experimental data, different mathematical models, based on integer order differential equations, have been proposed for better understanding [3] [4], control [5] [6], and early prediction of the glucose-insulin system [7] [8].

Fractional order derivatives and integrals provide generalization of differentiation and integration. Moreover, they incorporate memory and hereditary effects, because of which

they are more powerful and realistic in terms of representing a phenomenon. For the same reasons, fractional calculus has got much attraction in recent years, and is being applied in various fields, for example, viscoelasticity, neural dynamics, ecology, bioengineering, robotics, electronics, and control theory [9]. Modeling of anomalous diffusion, and viscoelastic materials are particular examples in this regard [10]. In literature, fractional order modeling of different diseases has also been done, opening a door for the deep understanding of their complex nature. Examples include fractional order models of influenza [11], dengue [12], malaria [13], and tuberculosis [14].

The contribution of this paper is to demonstrate the superiority of fractional order modeling over the integer order one. The most famous nonlinear Bergman's model for the description of glucose-insulin system in blood is modified to fractional order version. After that, it is shown through simulations and comparison of the data that the fractional order model gives better description of the system compared to the original integer order model.

In the next section, the three most important definitions of fractional derivatives are included. Section 3 briefly describes the blood glucose-insulin system, and diabetes mellitus. Section 4 presents different mathematical models. Section 5, through simulations, compares the models with the experimental data, and, finally, Section 6 concludes the paper.

II. BASIC DEFINITIONS

Represented by ${}_a\mathcal{D}_t^\alpha$, fractional derivative is given by

$$\begin{cases} \frac{d^\alpha}{dt^\alpha} & \mathbf{Re}(\alpha) > 0 \\ 1 & \mathbf{Re}(\alpha) = 0 \\ \int_a^t (d\tau)^{-\alpha} & \mathbf{Re}(\alpha) < 0 \end{cases}$$

where $\alpha \in \mathbb{R}$ in general, but may belong to the set of complex numbers [15]. Multiple definitions of fractional derivatives exist, each of which is mathematically correct, and has its own interpretation. Most important of them are three.

1. Grünwald-Letnikov Definition

First introduced, in 1867, by Anton Karl Grünwald, and then, in 1868, by Aleksey Letnikov [16], the definition over the interval $[a, t]$ is given by

$${}_a^{GL} \mathcal{D}_t^\alpha f(t) = \lim_{j \rightarrow 0} \frac{1}{j^\alpha} \sum_{n=0}^{\lfloor \frac{t-a}{j} \rfloor} (-1)^n \binom{\alpha}{n} f(t - nj)$$

where $n \in \mathbb{N}$, j is the step size, and binomial $\binom{\alpha}{n}$ can be computed using gamma function in the following way.

$$\binom{\alpha}{n} = \frac{\Gamma(1 + \alpha)}{\Gamma(1 + n)\Gamma(1 - n + \alpha)}$$

where gamma function is defined as

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx, \quad z \in \mathbb{C}$$

2. Riemann-Liouville Definition

Defined by Riemann, in 1847, the definition over the interval $[b, t]$ is given by

$${}_b^{RL} \mathcal{D}_t^\beta f(t) = \frac{1}{\Gamma(q - \beta)} \frac{d^q}{dt^q} \int_b^t \frac{f(T)}{(t - T)^{\beta - q + 1}} dT$$

where $q - 1 < \beta < q$, $\beta > 0$, $q \in \mathbb{N}$, and $\Gamma(\cdot)$ is the gamma function.

For $0 < \beta < 1$, the Riemann-Liouville (RL) operator is reduced to

$${}_b^{RL} \mathcal{D}_t^\alpha f(t) = \frac{1}{\Gamma(1 - \beta)} \frac{d}{dt} \int_b^t \frac{f(T)}{(t - T)^\beta} dT$$

3. Caputo Definition

In order to overcome the shortcomings of the RL operator, in 1967, Caputo [17] redefined the noninteger order derivative. The definition over the interval $[c, t]$ is given by

$${}_c^C \mathcal{D}_t^\gamma f(t) = \frac{1}{\Gamma(r - \gamma)} \int_c^t \frac{D^r f(T)}{(t - T)^{\gamma - r + 1}} dT$$

where $r - 1 \leq \gamma < r$, and D^r is the r^{th} derivative.

III. THE BLOOD GLUCOSE-INSULIN SYSTEM

One of the most famous examples of physiological systems of a human body is the system of glucose and insulin in blood. This system keeps the concentration of glucose in blood in the normal range which is (70-110 mg/dL). The terms ‘‘Hypoglycemia’’, and ‘‘Hyperglycemia’’ are used for the glucose level less than 70 mg/dL and greater than 180 mg/dL, respectively.

For a normal healthy person, when glucose level increases from the normal range, for example, after taking meal, pancreas receives a signal and starts secreting a hormone called insulin. The secretion of insulin results in lowering glucose level back to the normal range. Similarly, when glucose level decreases from the normal range, for example, after exercise, pancreas receives a signal but now, it releases glucagon which enforces the liver to discharge glucose because of which, the glucose level shifts back to the normal range.

If glucose level of a person decreases from the normal value, and remains there, it means that the person is suffering from a disease known as Diabetes Mellitus.

Type-1 and Type-2 are the most common types of diabetes. In case of former, pancreas is not able to secrete the insulin because the auto-immune system destroys the cells that are responsible for secreting the insulin. So, patients require regular injections of insulin. In case of latter, pancreas is able to secrete some insulin, but either it is not enough to lower down the glucose level or body resists its effect.

(1) Fasting Blood Glucose Test, (2) Oral Glucose Tolerance Test [18], and (3) Intravenous Glucose Tolerance Test [19] are the three most common tests for diabetes detection.

IV. MATHEMATICAL MODELS

For mathematical modeling, on the basis of experimental data, a hypothesis is made which is followed by a mathematical model that describes the phenomenon under study. Once a mathematical model is obtained, its accuracy is tested against new data. If model adequately describes the phenomenon for new data, then this model is ready to mathematically replace the phenomenon, otherwise, it is required to refine or replace that model with another one. Prediction of future performance of a system is one of the most important characteristics of a good model. A detailed discussion on modeling, simulation and control of physiological systems is given in [20]. Each model is correct under certain terms and conditions. Different mathematical models for the description of the system of glucose and insulin in blood are given below.

A. Linear Model

On the basis of the study of concentration of insulin and glucose, done on normal persons, a linear model is given in [4]. This model is given as

$$\begin{aligned} \frac{dG}{dt} &= -m_1 G(t) + m_2 I(t) + m_1 G_b \\ \frac{dX}{dt} &= -m_2 X(t) + m_3 I(t) - m_3 I_b + m_6 I_b \\ \frac{dI}{dt} &= -m_3 I(t) + m_4 G(t) + m_4 m_5 - m_6 I(t) + m_6 I_b \end{aligned} \quad (1)$$

where

$G(t)$: Concentration of glucose (mg/dL)
 $X(t)$: Effect of insulin (min^{-1})
 $I(t)$: Concentration of insulin ($\mu U/mL$)

G_b : Glucose's basal value (mg/dL)

I_b : Insulin's basal value ($\mu U/mL$)

Constants:

m_1 : Rate of glucose uptake, independent of insulin (min^{-1})

m_2 : Rate of decrease ability of tissue for glucose uptake (min^{-1})

m_3 : Increase in ability of tissue for glucose uptake per unit of I_b , independent of insulin ($(\mu U/mL)min^{-2}$)

m_4 : Rate of secretion of pancreatic β cells after dose of glucose with glucose level above system $[(\mu U/mL)(mg/dL)^{-1}min^{-2}]$

m_5 : The value of glucose on reaching which insulin is released.

m_6 : Rate of decay for insulin (min^{-1})

For a normal person, values of parameters, taken from [4], are given in the Table I.

This linear model has been modified into fractional order

Parameter	Value
m_1	0.0317000
m_2	0.0123
m_3	4.92×10^{-6}
m_4	0.0039
m_5	79.0353
m_6	0.2659
G_b	80
I_b	7

TABLE I: Parameters and their values

model in [21], where fractional order differential equations have been solved using Laplace Adomain Decomposition method. The model is as follows.

$$\begin{aligned}
{}_0\mathcal{D}_t^{\alpha_1} G(t) &= -m_1 G(t) + m_2 I(t) + m_1 G_b \\
{}_0\mathcal{D}_t^{\alpha_2} X(t) &= -m_2 X(t) + m_3 I(t) - m_3 I_b + m_6 I_b \\
{}_0\mathcal{D}_t^{\alpha_3} I(t) &= -m_3 I(t) + m_4 G(t) + m_4 m_5 - m_6 I(t) + m_6 I_b
\end{aligned} \tag{2}$$

$$0 < \alpha_1 \leq 1, \quad 0 < \alpha_2 \leq 1, \quad 0 < \alpha_3 \leq 1$$

B. Bergman's Minimal Model

Modeling of the system of glucose and insulin is pioneered by Richard N. Bergman et al [22]. Bergman proposed and then compared seven different models in order to find out the most accurate, and declared one out of the seven models as the best. This model is nonlinear with minimum number of parameters, and given the name "Bergman's Minimal Model". In this model, body is supposed to be a container with some preexisting concentration of glucose and insulin. For the same reason, this model is described as one compartment model. In fact, this model consists of two sub models, glucose minimal model and insulin minimal model [23]. Former describes the kinetics of glucose while later describes the kinetics of insulin. The model is given as

$$\begin{aligned}
\frac{dG}{dt} &= -(q_1 + X(t))G(t) + q_1 G_b \\
\frac{dX}{dt} &= -q_2 X(t) + q_3 (I(t) - I_b) \\
\frac{dI}{dt} &= -q_6 [G(t) - q_5]^+ t - q_4 (I(t) - I_b)
\end{aligned} \tag{3}$$

where

$G(t)$: Concentration of glucose (mg/dL)

$X(t)$: Effect of active insulin (min^{-1})

$I(t)$: Concentration of Insulin ($\mu U/mL$)

G_b : Glucose's basal value (mg/dL)

I_b : Insulin's basal value ($\mu U/mL$)

$[G(t) - q_5]^+$: $\max(G(t) - q_5, 0)$

Constants:

q_1 : Clearance rate of glucose without insulin (min^{-1})

q_2 : Active clearance rate due to insulin (min^{-1})

q_3 : Caused by insulin, ability for glucose uptake ($min^{-2}(L/mU)$)

q_4 : Rate of decay for insulin (min^{-1})

q_5 : The target value of glucose (mg/dL)

q_6 : Rate of secretion of pancreatic β cells after dose of glucose ($mUdLmin^{-1}/Lmg$)

S_G : q_1 : Glucose effectiveness

S_I : $\frac{q_3}{q_2}$: Insulin effectiveness

Values of the parameters, given in Table II, are taken from [23], in which these values are obtained using a computer program, named "MINMOD".

Parameter	Value
q_1	0.03082
q_2	0.02093
q_3	1.062×10^{-5}
q_4	0.3
q_5	94
q_6	0.3349×10^{-2}
G_b	92
I_b	7.3

TABLE II: Parameters and their values

Additionally, it has been shown in [23] that best curve fitting is obtained for $G(0) = 287 mg/dL$, $X(0) = 0 min^{-1}$, and $I(0) = 403.4 (\mu U/mL)$.

Integer order Bergman's model is replaced by fractional order Bergman's model in [24]. The existence and uniqueness of the modified model is discussed therein. The extended model is given as

$$\begin{aligned}
{}_0\mathcal{D}_t^\alpha G(t) &= -(q_1 + X(t))G(t) + q_1 G_b \\
{}_0\mathcal{D}_t^\beta X(t) &= -q_2 X(t) + q_3 (I(t) - I_b) \\
{}_0\mathcal{D}_t^\gamma I(t) &= -q_6 [G(t) - q_5]^+ t - q_4 (I(t) - I_b)
\end{aligned} \tag{4}$$

$$0 < \alpha \leq 1, \quad 0 < \beta \leq 1, \quad 0 < \gamma \leq 1$$

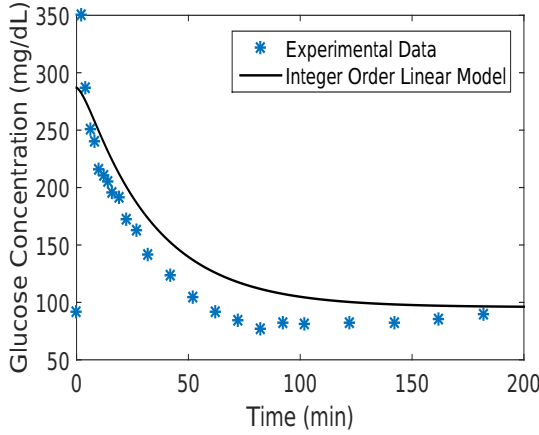
In this paper, we have utilized the above mentioned model, and tuned for the order of fractional derivative which ensures better fit. Furthermore, based upon the experimental data used by Bergman in [23], we demonstrate that, with the tuned fractional order derivatives, fractional order Bergman's minimal model is better representative of the system of glucose and insulin in blood as compared to its integer order version.

V. COMPARISON OF MODELS WITH EXPERIMENTAL DATA

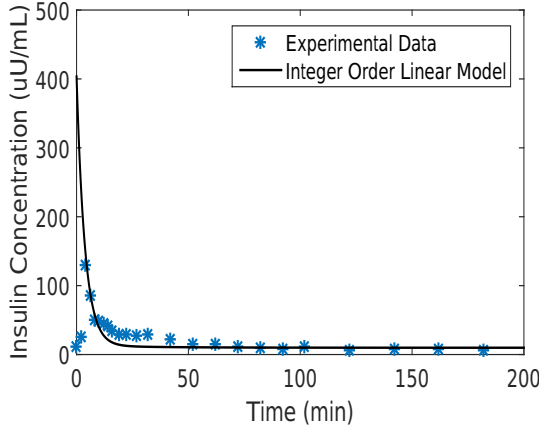
For comparison of models, experimental data given in [23] has been utilized.

A. Integer Order Linear Model

Integer order linear model is simulated in MATLAB using standard ODE45 solver. Values of parameters have been taken from Table I. Initial values taken are $(G(0), X(0), I(0)) = (287, 0, 403.4)$. Simulation results are shown in Fig. 1.



(a) Glucose concentration versus time



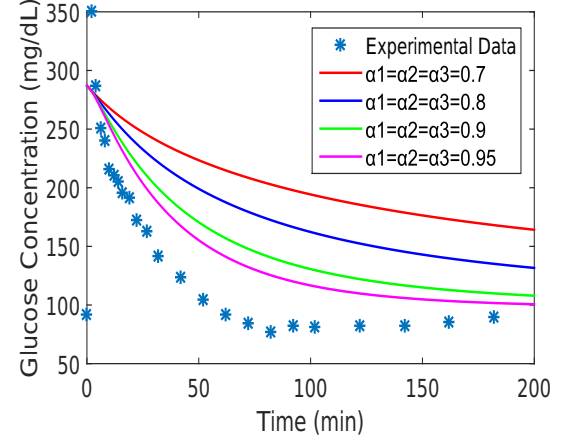
(b) Insulin concentration versus time

Fig. 1: Linear model (Integer order)

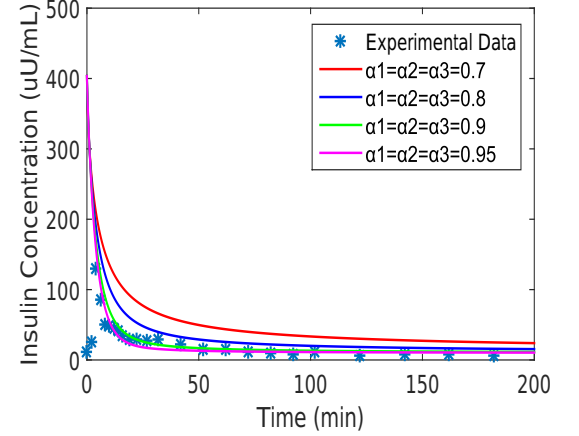
B. Fractional Order Linear Model

Fractional order linear model is simulated in MATLAB using Adams Bashforth Moulton type PECE method. Complete algorithm, and its properties are given in [25]. Values of

parameters have been taken from Table I. Initial values taken are $(G(0), X(0), I(0)) = (287, 0, 403.4)$. Simulation results are shown in Fig. 2.



(a) Glucose concentration versus time



(b) Insulin concentration versus time

Fig. 2: Linear model (Fractional order)

C. Bergman's Minimal Model

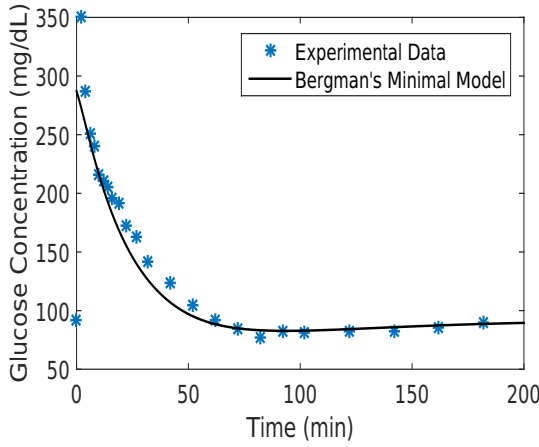
Bergman's minimal model is simulated in MATLAB using ODE15s solver. Values of parameters have been taken from Table II. Initial values taken are $(G(0), X(0), I(0)) = (287, 0, 403.4)$. Simulation results are shown in Fig. 3.

D. Modified Fractional Order Bergman's Minimal Model

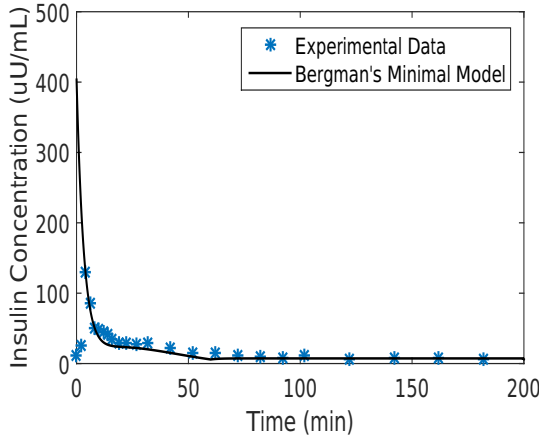
Modified fractional order version of the model is simulated in MATLAB using Adams Bashforth Moulton type PECE method, given in [25]. Values of parameters have been taken from Table II. Initial values taken are $(G(0), X(0), I(0)) = (287, 0, 403.4)$. Simulation results are shown in Fig. 4.

E. Integer Order Versus Fractional Order Bergman's Model

For integer order model, simulation is done using the values of parameters, which are best estimates, according to [23]. In case of modified fractional order model, there is provision



(a) Glucose concentration versus time



(b) Insulin concentration versus time

Fig. 3: Integer order Bergman's minimal model

of changing fractions α , β , γ as well as parameters p_1 to p_5 . Taking same parameter values that were used for original Bergman's model, and changing fractions in the way $\alpha=\beta=\gamma$, it can be observed that for $\alpha=\beta=\gamma=0.95$, better fit is obtained. Simulations of integer order and modified fractional order Bergman's minimal model are shown in Fig. 5.

F. Comparison Results

From the plots of integer order and fractional order linear models, it can be seen that these models are crude representation of the blood glucose-insulin system. Original Bergman's minimal model is integer order nonlinear model. This nonlinear model has been modified into fractional order model, and simulated for different fractions. For 0.95, this fractional order model gives better fit on the experimental data. It is worthy to note that the provision of changing fractions in different ways as well as changing parameters is still there, and by availing this provision, it is possible to get a very close fit. Comparison of average and rms values of absolute difference from the experimental data is given in the Table III.

Variable	Integer Order Model	Fractional Order Model $\alpha=\beta=\gamma=(0.95)$
Average Value		
Glucose Concentration	8.9	6.3
Insulin Concentration	5.86	3.1
rms Value		
Glucose Concentration	12.3	8.4
Insulin Concentration	7.45	4.95

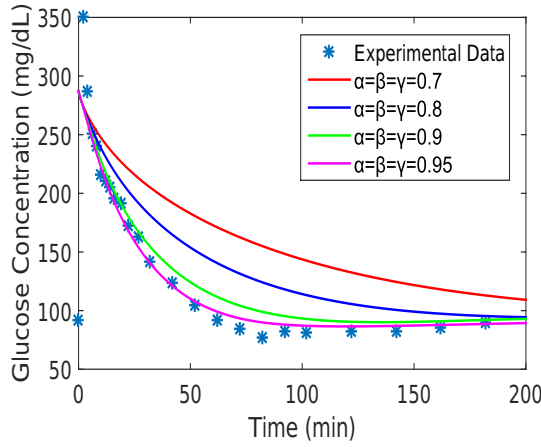
TABLE III: Average and rms values of absolute difference from the experimental data

VI. CONCLUSION

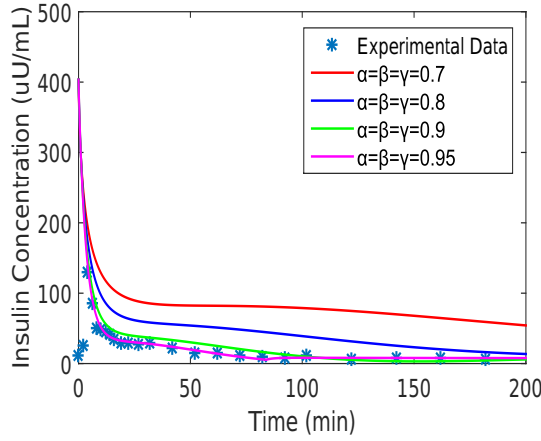
There are different linear and nonlinear mathematical models to describe the system of glucose and insulin in blood. The most famous nonlinear Bergman's minimal model is modified to fractional order model, and on the basis of simulations and comparison with the experimental data, it is concluded that the fractional order version of Bergman's minimal model is better representative of the system than its integer order form.

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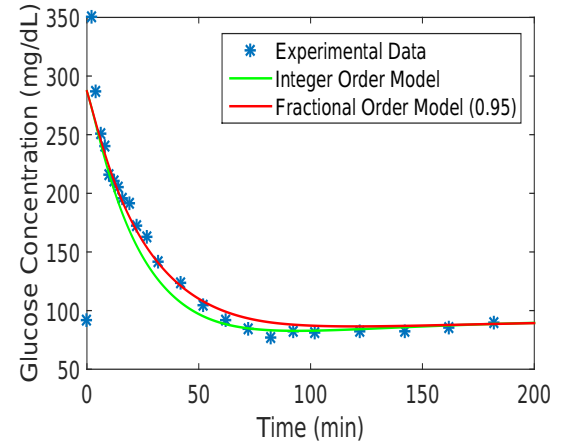


(a) Glucose concentration versus time

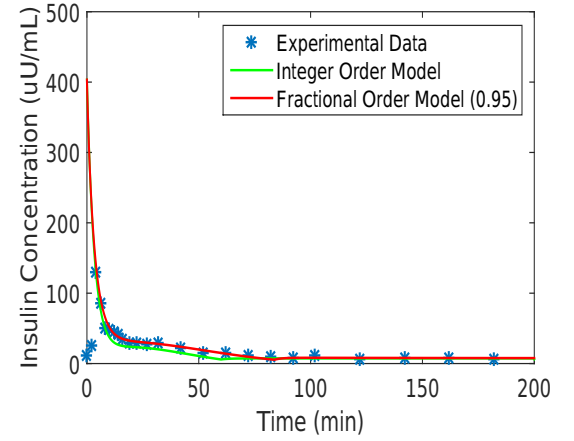


(b) Insulin concentration versus time

Fig. 4: Modified fractional order Bergman's minimal model



(a) Glucose concentration versus time



(b) Insulin concentration versus time

Fig. 5: Bergman's minimal model—(Fractional order as well as integer order)

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